i) NS2 Simulator – Introduction: NS2 is an open-source, discrete-event network simulator used to simulate and analyze network protocols.

ii) Number of Packets Dropped: The number of packets dropped during the network simulation is calculated from the trace file.

iii) Packets Dropped by TCP/UDP: The number of packets dropped by TCP and UDP protocols is identified from the simulation trace file.

iv) Packets Dropped Due to Congestion: Packets are dropped when network congestion occurs due to limited bandwidth or buffer capacity.

v) Compare Data Rate & Throughput: Data rate is the amount of data transmitted per unit time, whereas throughput is the amount of data successfully received per unit time.

Result: Thus, the NS2 simulator was studied, and packet drops, TCP/UDP packet drops, congestion-related packet drops, data rate, and throughput were analyzed.